Inside a Router: Architecture and Operation

1 Introduction

Routers are the core packet-switching devices of the Internet. After understanding the distinction between **routing** (control plane) and **forwarding** (data plane), we now explore what happens **inside** a router.

A router's internal architecture determines how fast and how efficiently packets can move from incoming links to outgoing links.

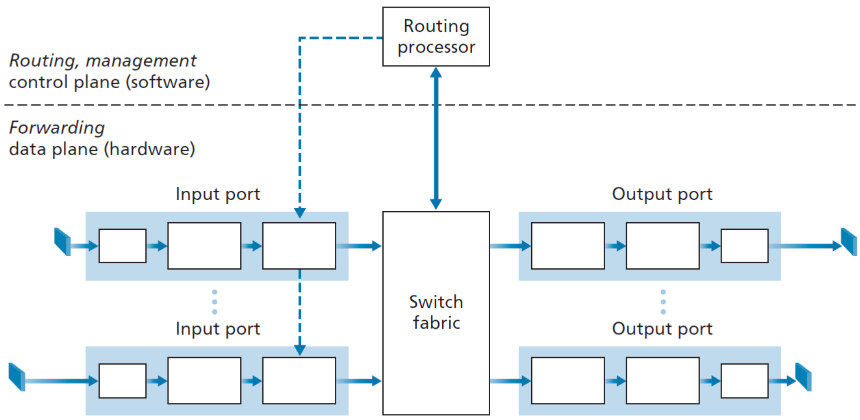

Figure 1: High-level router architecture showing input ports, switching fabric, output ports, and routing processor.

Routers have four primary components:

- **Input Ports**
- **Switching Fabric**
- **Output Ports**
- **Routing Processor**

Each contributes to high-speed forwarding and routing.

2 Input Ports

Input Port Responsibilities

- **Physical-layer processing** – terminates the incoming physical link.
- **Link-layer processing** – interoperates with the Layer-2 protocol.
- **Lookup (Forwarding) function** – checks forwarding table to find output port.
- **Control packet delivery** – sends routing/control packets to the routing processor.

Why Input Ports Must Be Hardware-Fast

At 100 Gbps, a new 64-byte packet arrives every:

$$5.12 \text{ nanoseconds}$$

Therefore input ports must be implemented entirely in hardware.

Real-World Example

The Juniper MX2020 supports:

- **800 × 100 Gbps ports**
- **800 Tbps total switching capacity**

Such speeds demand hardware-level processing, not software.

3 Switching Fabric

Switching Fabric

The switching fabric is the router's internal high-speed "mini-network" that moves packets from input ports to output ports.

Common Fabric Architectures

- **Memory-based switching** – oldest, slowest.
- **Bus-based switching** – all ports share a single bus.
- **Crossbar/interconnection networks** – parallel, scalable, high-speed.

Modern routers use crossbars or multi-stage networks to avoid internal bottlenecks.

4 Output Ports

Output Port Responsibilities

- Buffer packets arriving from the switching fabric.
- Apply link-layer encapsulation operations.
- Perform physical-layer transmission onto outgoing links.

Output Buffering and Congestion

When multiple packets simultaneously need the same output link, queueing occurs. This is a common location for:

- Packet loss
- Queueing delay
- Congestion

Thus output ports often contain large high-speed buffers.

5 Routing Processor

Routing Processor (Control Plane)

- Runs routing protocols (OSPF, RIP, BGP).
- Updates routing and forwarding tables.
- Manages link-state information.
- Handles network management (e.g., SNMP, telemetry).
- In SDN: communicates with remote SDN controller and installs flow rules.

Performance Contrast

- **Data plane:** nanoseconds (hardware)
- **Control plane:** milliseconds/seconds (software)

Thus forwarding is hardware-accelerated, but routing runs on a general-purpose CPU.

6 Hardware vs. Software Responsibilities

- **Hardware (fast path):** input ports, output ports, switching fabric.
- **Software (slow path):** routing protocols, management tasks.

Chipsets may be custom-designed or based on merchant silicon (Broadcom, Intel).

7 Roundabout Analogy

To visualize router behavior, imagine a roundabout:

Destination-Based Forwarding

A car:

- tells its final destination,
- attendant chooses correct exit,
- car enters and leaves at that exit.

This is the same as normal IP destination-based forwarding.

Generalized Forwarding

Exit choice may depend on:

- origin (source IP)
- vehicle type (protocol fields)
- roadworthiness (ACLs/firewalls)

This mirrors SDN / OpenFlow matching multiple header fields.

Where Congestion Occurs

- Too-fast arrivals → input blocking.
- Slow roundabout movement → fabric contention.
- Too many cars wanting the same exit → output queue buildup.

Exactly the same bottlenecks exist inside routers.

8 Conclusion

Routers combine:

- High-speed hardware for forwarding
- Flexible software for routing and management

Understanding these internals helps explain router performance, bottlenecks, and design trade-offs.